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Abstract

This ACM-like template, derived from the original `acmart` document class, is intended to support the preparation of scientific papers for the symposia and workshops that compose CBSOft. It preserves the visual identity and structural organization of the ACM consolidated format while adapting its elements to comply with the editorial and formatting guidelines adopted by the Brazilian Computer Society. This template can be used for both review and camera-ready versions, allowing authors to prepare their manuscripts with minimal structural changes. The abstract text must be written as a single continuous paragraph, without paragraph breaks.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

The article template adopted for CBSOft¹ is an ACM-like version derived from the original ACM consolidated template. It preserves the visual identity and structural organization of the `acmart` class, while adapting its elements to comply with the formatting guidelines and editorial context of CBSOft and the Brazilian Computer Society (SBC).

Originally introduced by ACM in 2017, the consolidated template was designed to provide a consistent $\text{\LaTeX}$ style across a wide range of publications and to incorporate accessibility and metadata-extraction features to support digital libraries. In this adapted version, the principles are maintained, while ACM-specific submission requirements have been removed or adjusted to align with the CBSOft publication process.

2 Template Overview

As noted in the introduction, this ACM-like template for CBSOft is derived from the original `acmart` document class and can be used to

prepare different types of submissions accepted by the conference, including full research papers, short papers, tool papers, and other publication formats defined by the CBSOft symposia and workshops. The template supports the main stages of the publication process, from review versions to final camera-ready copies, by selecting appropriate template styles and parameters.

This document presents the main features of the adapted document class and explains how to use its elements in accordance with the CBSOft formatting guidelines. Since this template is based on the ACM consolidated format, authors familiar with the original `acmart` class will find its structure and commands largely preserved, while ACM-specific requirements have been removed or adjusted to fit the CBSOft publication process.

2.1 Template Parameters

There are a number of *template parameters* which modify some part of the applied template style. A complete list of these parameters can be found in the *$\text{\LaTeX}$ User's Guide*.

Frequently-used parameters, or combinations of parameters, include:

- `anonymous, review`: Suitable for a double-anonymized conference submission. It anonymizes the work and includes line numbers.
- `authorversion`: Produces a version of the work suitable for posting by the author.
- `screen`: Produces colored hyperlinks.

3 Modifications

Modifying the template, including but not limited to adjusting margins, typeface sizes, line spacing, paragraph and list definitions, and the use of the `\vspace` command to manually adjust the vertical spacing between elements of your work, **is not allowed**.

Your document will be returned to you for revision if modifications are discovered.

¹CBSOft website: <https://cbsoft.sbc.org.br/>

4 Typefaces

The `acmart` document class requires the use of the Libertine typeface family. Your $\text{\TeX}$ installation should include this set of packages. Please do not replace it with other typefaces. The `lmodern` and `ltimes` packages should not be used, as they will override the built-in typeface families.

5 Title Information

The title of your work should use capital letters appropriately – <https://capitalizemytitle.com/> has useful rules for capitalization. Use the `title` command to define the title of your work. If your work has a subtitle, define it with the `subtitle` command. Do not insert line breaks in your title.

If your title is lengthy, you must define a short version to be used in the page headers to prevent overlapping text. The `title` command has a “short title” parameter:

```
\title[short title]{full title}
```

6 Authors and Affiliations

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors’ names should not be abbreviated; use full first names wherever possible. Inclusion of e-mail addresses for all authors is mandatory and must follow the font style defined in the template.

Immediately below the title, each author must be declared using the following format:

```
\author{FirstName Surname}
\affiliation{
  \institution{Department Name, Institution/University Name}
  \city{City}
  \country{Country}
}
```

Grouping authors’ names or e-mail addresses, or providing an “e-mail alias,” as shown below, is not acceptable:

```
\author{Brooke Aster, David Mehldau}
\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

For papers with multiple authors, the surnames of the authors must be included in the page headers. The list of author names shown in the header must not exceed the column width. Therefore, only the surnames should be used, preferably in the form “Surname et al.” when appropriate. This shortened list must be explicitly defined using the following command, placed immediately after the last `\author` declaration:

```
\renewcommand{\shortauthors}{Surname1stAuthor et al.}
```

Omitting this command may cause the full list of authors to be automatically used in the page headers, which can lead to text overlap and layout issues.

The `authornote` and `authornotemark` commands allow a note to apply to multiple authors. For example, if the first two authors of an article contributed equally to the work.

Before submitting the camera-ready version, authors must verify the following:

- All authors are declared individually using the required format.

Table 1: Frequency of special characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

- All e-mail addresses have been correctly included.
- The font used for e-mail addresses is consistent with the template.
- The shortened author list defined by `\shortauthors` contains only surnames and fits within the column width of the header.

7 Sectioning Commands

Your work should use standard $\text{\LaTeX}$ sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` should be numbered; do not remove the numbering from the commands.

All numbered section and subsection titles must follow the same capitalization style: the first letter of each word in the title must be capitalized (Title Case).

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed**.

Below are examples of sectioning commands:

7.1 Subsection

This is a subsection.

7.1.1 Subsubsection. This is a subsubsection.

Paragraph. This is a paragraph.

Subparagraph This is a subparagraph.

8 Tables

This ACM-like CBSOft template, based on the `acmart` document class, includes the `booktabs` package (<https://ctan.org/pkg/booktabs>) for preparing high-quality tables.

Table captions are placed **above** the table.

Because tables cannot be split across pages, the best placement is typically at the top of the page nearest their initial citation. To ensure this proper “floating” placement of tables, use the `table` environment to enclose the table’s contents and the table caption. The contents of the table must be placed in the `tabular` environment to ensure proper row and column alignment. Do not use horizontal lines between rows with `hline`, except for the table header and marking its end. Do not use vertical lines between columns in any case.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page’s live area, use the `table*` environment to enclose the table’s contents and the table caption. As with a single-column table, this

wide table will “float” to a location deemed more desirable. Immediately following this sentence is the point at which Table 2 is included in the input file. Once again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Always use `midrule` to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognize table headers and support their users in navigating tables more easily.

9 Math Equations

You may want to display math equations in three distinct styles: inline, numbered, or non-numbered display. Each of the three is discussed in the next sections.

9.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the `math` environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω , available in $\text{\LaTeX}$ [25]; this section will simply show a few examples of in-text equations in context. This is an equation in in-line math style:

$$\lim_{n \rightarrow \infty} x = 0,$$

It looks slightly different when set in display style (see next section).

9.2 Display Equations

A numbered display equation, set off by vertical space from the text and centered horizontally, is produced by the `equation` environment. An unnumbered display equation is produced by the `displaymath` environment.

Once again, in either environment, you can use any of the symbols and structures available in $\text{\LaTeX}$; this section gives just a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n \rightarrow \infty} x = 0 \quad (1)$$

Notice how it is formatted somewhat differently in the `displaymath` environment. Now, we will enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

And this is another numbered equation:

$$\sum_{i=0}^{\infty} x_i = \int_0^{\pi+2} f \quad (2)$$

10 Figures

The `figure` environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Figure captions are placed **below** the figure.

Every figure should also have a figure description with the `caption` command unless it is purely decorative. These descriptions convey what



Figure 1: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

is s in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

11 Citations and Bibliographies

The use of $\text{\LaTeX}$ for the preparation and formatting of one’s references is strongly recommended. Authors’ names should be complete. Use full first names (“Donald E. Knuth”) instead of initials (“D. E. Knuth”), and the identifying features of a reference must be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

`bibfile` is the $\text{\LaTeX}$ filename without the `.bib` suffix.

Citations and references are numbered by default. Some examples:

- A paginated journal article [2]
- An enumerated journal article [11]
- A reference to an entire issue [10]
- A monograph (whole book) [24]
- A monograph/whole book in a series [18]
- A divisible book such as an anthology or compilation [13]
- A divisible book (series not included since no volume number is given) [14]
- A chapter in a divisible book [37]
- A chapter in a divisible book in a series [12]
- A multi-volume work as a book [23]
- Paginated proceedings articles [3, 16]
- A proceedings article with all possible elements [36]
- An enumerated proceedings article [15]
- An informally published work [17]
- Preprints [6, 8]

Table 2: Some typical commands

Command	A Number	Comments
<code>\author</code>	100	Author
<code>\table</code>	300	For tables
<code>\table*</code>	400	For wider tables

- A doctoral dissertation [9]
- A master’s thesis [4]
- Online documents / World Wide Web resources [1, 29, 38]
- Video games (Case 1) [28]
- Video games (Case 2) [26, 27]
- A patent (Case 3) [35]
- Work accepted for publication [32]
- ‘YYYYb’ test for prolific author [33, 34]
- Citations containing duplicate DOI and URLs (e.g., some SIAM articles) [22]
- Multi-volume works as books [19, 20]
- A presentation [31]
- An article under review [7]
- Citations with DOIs [21, 22]
- Online citations [38–40]
- Artifacts [5, 30]

12 Recommendations for Appendices

If your work needs an appendix, add it before the `\end{document}` command at the conclusion of your source document.

Start the appendix with the appendix command:

```
\appendix
```

In the appendix, sections are lettered rather than numbered. This document has two appendices that demonstrate the section and subsection identification method.

13 Conclusion and Future Work

<text included by the authors>

Artifact Availability

This section must be included immediately after the Conclusion and before the Acknowledgments section. It must not be numbered and must be titled exactly as **Artifact Availability**.

If the paper provides research artifacts (such as datasets, source code, scripts, models, or experimental packages), the authors must describe what is available, where it can be accessed, and under which license or usage conditions.

If no artifacts are available, this section must still be included, explicitly stating that no artifacts are provided.

For papers written in Portuguese, this section must be titled **Disponibilidade de Artefatos**.

Acknowledgement

<text included by the authors>

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A Appendix Example 1

A.1 Part One

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi malesuada, quam in pulvinar varius, metus nunc fermentum urna, id sollicitudin purus odio sit amet enim. Aliquam ullamcorper eu ipsum vel mollis. Curabitur quis dictum nisl. Phasellus vel semper risus, et lacinia dolor. Integer ultricies commodo sem nec semper.

A.2 Part Two

Etiam commodo feugiat nisl pulvinar pellentesque. Etiam auctor sodales ligula, non varius nibh pulvinar semper. Suspendisse nec lectus non ipsum convallis congue hendrerit vitae sapien. Donec at laoreet eros. Vivamus non purus placerat, scelerisque diam eu, cursus ante. Etiam aliquam tortor auctor efficitur mattis.

B Appendix Example 2

Nam id fermentum dui. Suspendisse sagittis tortor a nulla mollis, in pulvinar ex pretium. Sed interdum orci quis metus euismod, et sagittis enim maximus. Vestibulum gravida massa ut felis suscipit congue. Quisque mattis elit a risus ultrices commodo venenatis eget dui. Etiam sagittis eleifend elementum.

Nam interdum magna at lectus dignissim, ac dignissim lorem rhoncus. Maecenas eu arcu ac neque placerat aliquam. Nunc pulvinar massa et mattis lacinia.